

Work sheet # 08

∴ Noman Ahmed

∴ 0863

∴ section "A" "AI"

Parabola ∴

Q: 1 → Draw the graphs of parabola

⑨ $y^2 = 10x$

Sol:-

Right cup parabola $a > 0$

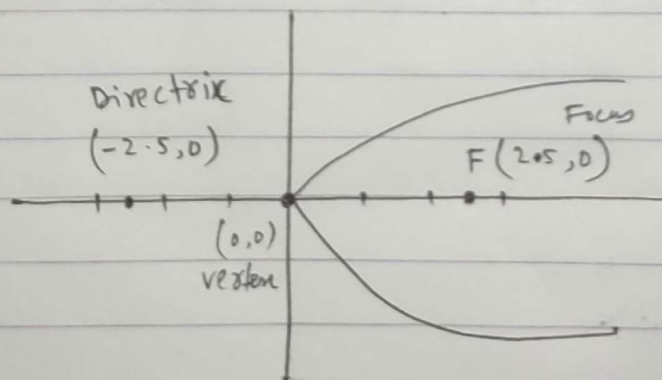
$$y^2 = 4ax$$

given equation $y^2 = 10x$

$$4a = 10$$

$$a = \frac{10}{4}$$

$$a = 2.5$$



b) $x^2 = -12y$

Sol.:

Down Cup parabola $a < 0$

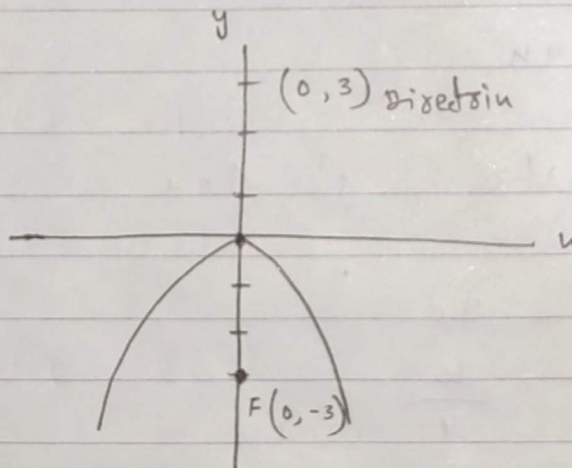
$$x^2 = 4ay \quad a < 0$$

Given equation $x^2 = -12y$

$$4a = -12$$

$$a = \frac{-12}{4}$$

$$a = -3$$



Q. 2 $\rightarrow$ Determine vertex, focus, latus rectum and equation of directrix of the following.

⑨ $y^2 = -8x$

Sol.

Left cup parabola $a < 0$

$$y^2 = 4ax$$

given equation, $y^2 = -8x$

$$4a = -8$$

$$a = \frac{-8}{4}$$

$$a = -2$$

- Focus $(-a, 0)$, $F(-2, 0)$
- Directrix $(a, 0)$, Directrix $(2, 0)$
- Vertex $(0, 0)$
- Latus rectum $= |4a| = 4 \times 2 = 8$

b) $x^2 = -16y$

soln

Down cup parabola $a < 0$

$$x^2 = 4ay \quad a < 0$$

given equation $x^2 = -16y$

$$4a = -16$$

$$a = \frac{-16}{4}$$

$$a = -4$$

• Focus $(0, -a)$ $F(0, -4)$

• Directrix $(0, a)$ $Directrix(0, 4)$

• Vertex $(0, 0)$

• Latus Rectum $|4a| = 4 \times 4 = 16$

③ $y^2 = 16u$

Solⁿ

Right up parabola $a > 0$

$$y^2 = 4au$$

Given equation $y^2 = 16u$

$$4a = 16$$

$$a = \frac{16}{4}$$

$$\boxed{a = 4}$$

- Focus $(4, 0)$, $F(4, 0)$
- Directrix $(-4, 0)$, Directrix $(-4, 0)$
- Vertex $(0, 0)$,
- Latus Rectum ~~16~~ $|4a| = 4 \times 4 = 16$

d) ~~Equation~~ $x^2 = 8y$

Sol.:

Up Cup parabola $a > 0$

$$x^2 = 4ay$$

Given equation $x^2 = 8y$

$$4a = 8$$

$$a = \frac{8}{4}$$

$$a = 2$$

• Focus $(0, a)$ • $F(0, 2)$

• Directrix $(0, -a)$ • Directrix $(0, -2)$

• Vertex $(0, 0)$

• Latus Rectum ~~Equation~~ $|4a| = 4 \times 2 = 8$

Q: 3 → Find eq: of parabola whose Focus and vertex are as under.

a) Vertex $(0,0)$: Focus $(5,0)$

Sol:

Right up parabola

$$y^2 = 4ax \quad a > 0$$

$$F(a, 0) \quad , \quad F(5, 0)$$

$$y^2 = 4ax$$

$$y^2 = 4 \times 5x$$

$$\boxed{y^2 = 20x} \quad \underline{\text{Ans.}}$$

b) Vertex $(0,0)$: Focus $(0,-2)$

Sol:

Down up parabola

$$x^2 = 4ay \quad a < 0$$

$$\text{Focus } (0, -a) \quad F(0, -2)$$

$$x^2 = 4ay$$

$$x^2 = 4 \times -2 \times y$$

$$x^2 = -8y$$

Q.4 → Find eq. of parabola whose vertex and point are given

① Vertex $(0,0)$; point $(3,4)$

Sol:-

Case: 1 up cup parabola

$$x = 3 \quad , \quad y = 4$$

$$x^2 = 4ay$$

$$(3)^2 = 4a \times 4$$

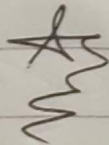
$$9 = 4a \times 4$$

$$a = \frac{9}{16}$$

$$x^2 = 4ay$$

$$x^2 = 4 \times \frac{9}{16} y$$

$$x^2 = \frac{9}{4} y$$



Case : 2

Right up parabola

$$x = 3, y = 4$$

$$y^2 = 4ax$$

$$(4)^2 = 4a \times 3$$

$$16 = 4a \times 3$$

$$\frac{16}{12} = a$$

$$a = \frac{4}{3}$$

$$y^2 = 4ax$$

$$y^2 = 4 \times \frac{4}{3} x$$

$$y^2 = \frac{16}{3} x$$

$$f_1$$

b) vertex $(0,0)$; point $(4,6)$

Case 1:

Cup - Up parabola

$$n = 4, y = 6$$

$$n^2 = 4ay$$

$$(4)^2 = 4a \times 6$$

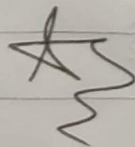
$$\begin{array}{r} 2 \quad 4 \quad 16 \\ 24 \overline{) 16} \\ \underline{12} \\ 4 \end{array} = \frac{2}{3}$$

$$a = \frac{2}{3}$$

$$n^2 = 4ay$$

$$n^2 = 4 \times \frac{2}{3} y$$

$$n^2 = \frac{8}{3} y$$



Case 2

Right cup parabola

$$u = 4, \quad y = 6$$

$$y^2 = 4ux$$

$$(6)^2 = 4 \times 4 \times x$$

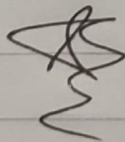
$$36 = 16x$$

$$x = \frac{9}{4}$$

$$y^2 = 4ux$$

$$y^2 = 4 \times \frac{9}{4} \times u$$

$$y^2 = 9u$$



→ Ellipse ::

Exp :: 01 ::

Find the Foci, vertices, co-vertices, latus rectum and eq. of directorin of the ellipse.

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

Solr ::

Ellipse when major axis on x-axis

$$a^2 = 25$$

$$a = \pm 5$$

$$b^2 = 9$$

$$b = \pm 3$$

$$a > b$$

then

$$a^2 = b^2 + c^2$$

$$25 = 9 + c^2$$

$$25 - 9 = c^2$$

$$16 = c^2$$

$$\boxed{c = \pm 4}$$

Sol.

- Foci $(\pm c, 0)$: $F(\pm 4, 0)$
- Vertices $(\pm a, 0)$: vertices $(\pm 5, 0)$
- Co-vertices $(0, \pm b)$: co-vertices $(0, \pm 3)$
- Latus rectum $= \frac{2b^2}{a} = \frac{2 \times 9}{5} = \frac{18}{5}$

equation of directrix

$$n = \frac{a}{e}$$

$$e = \frac{c}{a}$$

$$n = \frac{5}{4/5}$$

$$n = \frac{5 \times 5}{4}$$

$$n = \frac{25}{4}$$

Example :: 02 → Find the equation of

ellipse with center at origin, whose major and minor axis are 10 and 8 units respectively and major axis is along x-axis.

Solution:-

When major axis is along x-axis then equation is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

major axis = 10 units

$$2a = 10$$

$$a = 5$$

minor axis = 8 units

$$2b = 8$$

$$b = 4$$

then equation of ellipse

$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

Example : 03 → Find the equation of ellipse passing through $(1, \sqrt{2})$ and $(\frac{\sqrt{6}}{2}, 1)$ where as the center is at origin and major axis is y-axis

Solution:-

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

• $(1, \sqrt{2})$ lies on the ellipse

$$\frac{(1)^2}{b^2} + \frac{(\sqrt{2})^2}{a^2} = 1$$

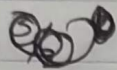
$$\boxed{\frac{1}{b^2} + \frac{2}{a^2} = 1} \rightarrow \text{eq (1)}$$

• $(\frac{\sqrt{6}}{2}, 1)$ lies on the ellipse

$$\frac{(\frac{\sqrt{6}}{2})^2}{b^2} + \frac{(1)^2}{a^2} = 1$$

$$\boxed{\frac{3}{2b^2} + \frac{1}{a^2} = 1} \rightarrow \text{eq (2)}$$

→ Multiply $\frac{3}{2}$ with equation 1 and then subtract from equation of eq. 2



$$\frac{3}{2b^2} + \frac{6}{2q^2} = \frac{3}{2}$$

$$\begin{array}{r} \frac{3}{2b^2} + \frac{6}{2q^2} = \frac{3}{2} \\ - \frac{3}{4b^2} + \frac{1}{q^2} = 1 \\ \hline \end{array}$$

$$\frac{3}{4b^2} - \frac{1}{q^2} = \frac{1}{2}$$

$$\frac{1}{q^2} (3 - 1) = \frac{1}{2}$$

$$(3 - 1) = \frac{q^2}{2}$$

$$2 = \frac{q^2}{2}$$

$$4 = q^2$$

$$\boxed{q = 2}$$

put in equation 1

$$\frac{1}{b^2} + \frac{2}{4^2} = 1$$

$$\frac{1}{b^2} + \frac{2}{(2)^2} = 1$$

$$\frac{1}{b^2} + \frac{2}{4} = 1$$

$$\frac{1}{b^2} + \frac{1}{2} = 1$$

$$\frac{1}{b^2} = 1 - \frac{1}{2}$$

$$\frac{1}{b^2} = \frac{2-1}{2}$$

$$\cancel{\frac{1}{b^2}} = \cancel{\frac{1}{2}}$$

$$b^2 = 2$$

$$b = \pm \sqrt{2}$$

equation of ellipse

$$\boxed{\frac{x^2}{2} + \frac{y^2}{4} = 1}$$

Example: 04 → Find the equation of ellipse with center at origin whose focus and vertex are $(8,0)$ and $(10,0)$ respectively.

Solution:

$$F(\pm c, 0) = (8, 0)$$

$$V(\pm a, 0) = (10, 0)$$

along x -axis

$$a = 10$$

$$c = 8$$

$$a^2 = 100$$

$$c^2 = 64$$

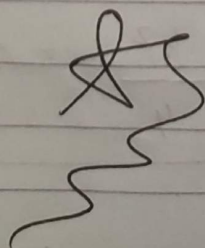
$$a^2 = b^2 + c^2$$

$$a^2 - c^2 = b^2$$

$$100 - 64 = b^2$$

$$36 = b^2$$

$$\boxed{\frac{x^2}{100} + \frac{y^2}{36} = 1} \rightarrow \text{equation of Ellipse}$$



Example: 05 $\Rightarrow$ Find the equation of ellipse with center at origin such that its focus is $(0, 3)$ and Latus rectum is of $\frac{32}{5}$ units

solution

Ellipse along y-axis

$$F(0, \pm c) = F(0, 3)$$

$$c = 3$$

$$c^2 = 9$$

$$\text{Latus rectum } \frac{2b^2}{a} = \frac{32}{5}$$

$$\text{then } a = 5$$

$$a^2 = 25$$

$$a^2 = b^2 + c^2$$

$$25 = b^2 + 9$$

$$25 - 9 = b^2$$

$$b^2 = 16$$

Equation of ellipse

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$\frac{x^2}{16} + \frac{y^2}{25} = 1$$

Example: 06 → Find the equation of an ellipse whose focus is $(5, 0)$ and minor axis is 12 units long and along y-axis where center is at origin.

Sol:-

$$F(c, 0) = (5, 0)$$

$$c = 5$$

$$c^2 = 25$$

$$\text{minor axis} = 12 \text{ units}$$

$$2b = 12$$

$$b = 6$$

$$b^2 = 36$$

$$a^2 = b^2 + c^2$$

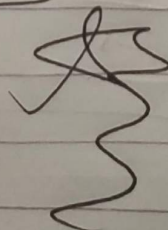
$$a^2 = 36 + 25$$

$$a^2 = 61$$

so equation of ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{61} + \frac{y^2}{36} = 1$$



Example : 07 → Find the equation of an ellipse whose minor axis is 10 units long and along x-axis where as latus rectum is 8 units long and center is at origin.

Sol:

along x-axis

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{minor axis} = 10$$

$$2b = 10$$

$$b = 5$$

$$b^2 = 25$$

$$\text{Latus rectum} = 8 \text{ units}$$

$$\frac{2b^2}{a} = 8$$

$$\frac{2 \times 25}{a} = 8$$

$$\frac{25}{84} = 9$$

$$a = \frac{25}{4} \Rightarrow a^2 = \left(\frac{25}{4}\right)^2$$

Equation of ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{\left(\frac{25}{4}\right)^2} + \frac{y^2}{25} = 1$$

Hyperbols

Example : 01 → Find eccentricity, foci,

ends of conjugate axis and
Latus rectum of hyperbola.

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

Sol:

along x-axis

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}$$

$$a^2 = 16 \rightarrow a = \pm 4$$

$$b^2 = 9 \rightarrow b = \pm 3$$

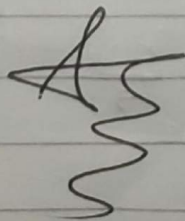
$$c^2 = a^2 + b^2$$

$$c^2 = 16 + 9$$

$$c^2 = 25$$

$$c = \pm 5$$

- Focus $(\pm c, 0)$; $F(\pm 5, 0)$
- V $(\pm a, 0)$; $V(\pm 4, 0)$
- $L.R = \frac{2b^2}{a} = \frac{2 \times 9}{4} = \frac{18}{4} = \frac{9}{2} = 4.5$
- eccentricity $e = \frac{c}{a} = \frac{5}{4}$
- End of conjugate axis are $(0, \pm bi)$
 $(0, \pm 3i)$
- Directrix = $\frac{a}{e}$ $e = \frac{c}{a}$
 $= \frac{4}{5/4} = 16/5$
- Asymptote $\Rightarrow y = \pm \frac{b}{a} x$
 $y = \pm \frac{3}{4} x$



Example : 02 Find the eccentricity

• equation of director,
eq. of Asymptote of Hyperbola

$$\frac{y^2}{9} - \frac{x^2}{4} = 1$$

Sol:-

along y-axis

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$a^2 = 9 \Rightarrow a = \pm 3$$

$$b^2 = 4 \Rightarrow b = \pm 2$$

~~$$c^2 = a^2 + b^2$$~~

$$c^2 = 9 + 4$$

$$c^2 = 13$$

$$c = \pm \sqrt{13}$$

Eccentricity $e = \frac{c}{a} = \boxed{\frac{\sqrt{13}}{3}}$

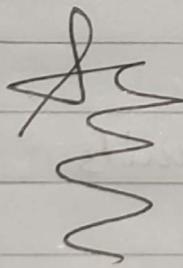
• Directrix $y = \pm \frac{a}{e} \quad e = \frac{c}{a}$

$$y = \pm \frac{3}{\frac{\sqrt{3}}{3}}$$

$$y = \pm \frac{9}{\sqrt{3}}$$

• Asymptote $y = \pm \frac{a}{b} x$

$$y = \pm \frac{3}{2} x$$



Example 03 → Find the equation of hyperbola If transverse axis and conjugate axis are 8 and 6 units long respectively, where center is at origin and transverse axis is along y-axis

Sol:-

along y-axis

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

transverse axis = $2a$

given value = 8 units

$$2a = 8$$

$$a = 4$$

$$a^2 = 16$$

conjugate axis = $2b$

given value = 6 units

$$2b = 6$$

$$b = 3$$

$$b^2 = 9$$

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so equation of hyperbola is

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$\boxed{\frac{y^2}{16} - \frac{x^2}{9} = 1}$$

~~Ans~~